



Science

Stage 8

Paper 1

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1	nutrient function	3	all four correct = 3 marks two or three correct = 2 marks one correct = 1 mark more than one line from a nutrient to different functions = 0 marks for that nutrient

Question	Answer	Marks	Further Information
2(a)	nucleus	1	
2(b)	electron	1	
2(c)	negative / –ve	1	Accept –1 / 1–

Question	Answer	Marks	Further Information
3(a)	exothermic	1	
3(b)	(idea of) highest temperature change	1	Accept largest temperature increase / biggest temperature rise Ignore the temperature change is 25°C

Question	Answer	Marks	Further Information
4(a)	(distance) tape measure / metre ruler (time) stopclock / stopwatch	2	Accept ruler Accept timer / chronometer / chronograph / clock with a second hand
4(b)	divide distance by time or distance ÷ time	1	Accept <u>distance</u> time Accept (s =) d/t
4(c)	(support his prediction) because the result for 30 degrees is more than the result for 20 degrees/10 degrees or because the result for 20 degrees/10 degrees is anomalous (do not support his prediction) because the result for 20 degrees is less than the result for 10 degrees	2	

Question	Answer	Marks	Further Information
5(a)	thermometer	1	
5(b)	line graph	1	Note graph is not sufficient Accept bar chart
5(c)	Mia (no mark) because climate is average temperature / weather is temperature in a particular place because climate is change over a long period of time / weather is temperature at a particular time	2	if Lily given = 0 marks for the question each correct explanation = 1 mark

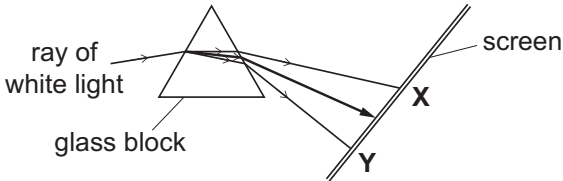
Question	Answer	Marks	Further Information
6(a)	C	1	more than one letter circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
6(b)	mitochondria	1	Accept mitochondrion
6(c)	(releases) energy	1	

Question	Answer	Marks	Further Information
7(a)	iron	1	Accept steel / nickel / cobalt
7(b)	increase number of cells	1	Accept increase the current / increase the voltage

Question	Answer	Marks	Further Information
8(a)	any two from (idea that) temperature (change) is (broadly) similar at the start (idea that) temperature (change) increases since 1950 / temperature (change) higher since 1950 (idea that) there are still peaks and troughs / temperature (change) is still going up and down at different times	2	Accept at start the temperature does not really change / at start the temperature is fairly constant Accept any dates between 1930 and 1950
8(b)	carbon dioxide	1	Accept CO ₂ / methane / CH ₄ / sulfur (VI) fluoride / SF ₆
8(c)	1.0 to 1.5 (°C)	1	Accept range of temperatures if within 1.0 to 1.5 (°C)

Question	Answer	Marks	Further Information
9(a)	respiratory	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
9(b)	a model (for a scientific concept or idea)	1	
9(c)	any one from the balloons represent the lungs the rubber sheet represents the diaphragm the tubing represents the trachea/bronchi the plastic bottle represents the ribcage	1	Accept an explanation about which part of the model relates to each part of respiratory system Accept (give) similarities Accept (give) differences Accept (show) how the respiratory system works

Question	Answer	Marks	Further Information
10(a)	30 (g / 100 cm ³)	1	
10(b)	B smallest mass of sugar in 100 cm ³	2	each correct answer = 1 mark Accept has the lowest concentration

Question	Answer	Marks	Further Information
11(a)	dispersion	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
11(b)	 ray of white light glass block screen X Y	1	line drawn between paths of red and violet with at least one correct arrow = 1 mark Note the line must be straight as judged by eye
11(c)	faster / greater / higher	1	Note different is not sufficient
Question	Answer	Marks	Further Information
12(a)	asteroids	1	
12(b)	rocks	1	

Question	Answer	Marks	Further Information
13(a)	decreases / gets smaller increases / gets larger increases / gets larger	2	all three correct = 2 marks one or two correct = 1 mark
13(b)	any two from grey squirrels out competed red squirrels for food grey squirrels out competed red squirrels for shelter grey squirrels introduced disease	2	each correct answer = 1 mark Accept grey squirrels hunted red squirrels / grey squirrels killed red squirrels as an extra marking point

Question	Answer	Marks	Further Information
14(a)	any two from same volume of acid same concentration of acid same mass of metal powder same temperature	2	Accept amount of metal as an alternative to mass Accept amount of acid if volume of acid and/or concentration of acid has not been given as an answer Accept same surface area of metal Ignore same equipment
14(b)(i)	65 (seconds)	1	
14(b)(ii)	calcium	1	
14(c)	zinc + (dilute) hydrochloric acid $\rightarrow$ zinc chloride + hydrogen	1	Accept = instead of $\rightarrow$ Note reactants can be either order Note products can be either order
14(d)	inert	1	

Question	Answer	Marks	Further Information
15(a)	(idea of gas) particles colliding with the bottle	1	Accept particles hit the bottle / particles bounce off the bottle Accept reference to wall rather than bottle Ignore collisions between particles Do not accept particles expand
15(b)	because particles move more slowly / particles hit sides of bottle less hard / fewer collisions with bottle / particles have less energy	1	Accept less particles collide with bottle (per unit time) / particles lose energy / particles collide less often with bottle / reduced collision frequency with bottle
15(c)	diffusion	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

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